

Chapter 7

1. Which members of the Circle class are encapsulated

The members of the Circle class that are encapsulated are anything that has a private method. If it has a public method, then it is not encapsulated.

2. What name must the constructor of a class have?

A constructor must have the same name as the class.

3. Explain the difference between the private and public access modifiers.

A private class is only accessible within the same class. This is usually used for encapsulation to protect data. A public class can be accessible anywhere.

4. Consider the following code. Is the last statement valid or invalid? Explain.

```
Circle dot = new Circle(2);  
dot.radius =5;
```

The last statement is only valid if the radius is public. If it is private, then the last statement would be invalid because the private method cannot be used in other classes.

5. Use the following class to answer the questions below:

```
public class Roo {  
  
    private int x;  
  
    public Roo {  
        x = 1;  
    }  
  
    public void setX(int z) {  
        x = z;  
    }  
  
    public int getX() {  
        return(x);  
    }  
  
    public int calculate() {
```

```

        x = x * factor();
        return(x);
    }

    private int factor() {
        return(0.12);
    }
}

```

a) What is the name of the class?

The name of the class is **Roo**

b) What is the name of the data member?

The name of the data member is **x**

c) List the accessor method.

The accessor method is **getX()**

d) List the modifier method.

The modifier method is **setX(int z)**

e) List the helper method.

The helper method is **factor()**

f) What is the name of the constructor?

The name of the constructor is **Roo()**

g) How many method numbers are there

The number of methods that are in the code is **getX(), setX(int z), factor(), and calculate()**.

6. What is the difference between a class and an object?

The difference between a class and an object is that a class is a data type that defines the variables for the state of an object. An object can store data and can perform actions and provide communication.

9. Use the following class data member definitions to answer the questions below:

```

public class Moo {
    private double y ;
    private static int x;
    private static final z;
}

```

a) Which data member is a constant?

The data member that is constant is z because of final

b) Which data members are variables?

The data members that are variables are y and x.

c) Which data member(s) are instance members?

The data member that is instance members is y because it is not static.

d) Which data member(s) are class members?

The data member that is class members is x and z because they are static

11. Compare and contrast overriding methods to overloading methods.

With overriding methods, it happens between two classes. However with overloading methods it happens within the class. Furthermore, overloading and overriding methods have the same name, but overloading have different parameters, and overriding has a return type. Additionally, overriding methods change the behaviour in a subclass whereas an overloading method, it provides many ways to call a method.